

Shahriyar Shawon

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Education

Drexel University

Bachelor of Science in Computer Science

Philadelphia, PA

September 2020 – June 2025

Concentrations: Operating Systems and Computational Imaging

Experience

Susquehanna International Group

Bala Cynwyd, PA

Software Engineer Co-op

SFG Team | C#, .NET Framework, MySQL, OracleDB

September 2023 – March 2024

- Enhanced data cohesion and accuracy of stock data by developing an API using **C#** and **.NET** to aggregate stock data from 15 **MySQL** and **OracleDB** sources
- Decreased redundant data fetching by 93% by redesigning a web app to use a central API
- Reduced stock information lookup time by 95% by introducing a caching mechanism
- Safeguarded privileged data by implementing a role-based access control system

MDF Metadata Team | Django, Python, Linux, MySQL, PostgreSQL

September 2022 – March 2023

- Improved data accuracy and operational reliability by building an alerting system using **Python** and **PostgreSQL** to monitor file delivery integrity and timeliness
- Reduced alert configuration time for developers by 50% by creating a CLI tool that streamlined setup
- Improved a file-user identification tool by implementing multiple detection methods, resulting in more reliable attribution of file usage

CorpSys Team | C#, .NET, MicrosoftSQL, EntityFramework

September 2021 – August 2022

- Enriched an internal project management tool by aggregating developer, stakeholder, pipeline, and environment data to improve team collaboration and project tracking
- Improved query performance by 20% by reducing the use of bridge tables in **MicrosoftSQL** by using LINQ with **EntityFramework**
- Developed the front-end interface for an API to allow general-purpose alerts to be created

Projects

Intro Detection for TV Shows | Python, OpenCV

- Developed an automated tool using **Python** and **OpenCV** to identify the end of TV show intros by comparing video frames to a known reference frame
- Applied Structural Similarity Index (SSIM) to detect a sharp drop in frame similarity, signaling end of the intro
- Leveraged ffmpeg for video metadata to calculate accurate start and end timestamps for intro segments

Dynamic Image Compression Engine | Python, C

- Built firmware in **C** for two RP2040 chips to interface with FPGA, IO Modules, and Power Monitor Chips
- Implemented custom image format to store lossy and lossless blocks of data using lossy and lossless codecs
- Created a desktop app to control camera settings on the fly and log power/data metrics from hardware
- Engineered a pipeline to compare software-based encoding performance to hardware based encoding

DragonFly3 Auto Connect Script | Bash, Linux

- Reduced time to connect to Drexel WiFi by 90% by automating creation of an Enterprise WiFi profile
- Significantly reduced **Linux** WiFi setup time by developing a **Bash** script now used by Drexel's College of Computing and Informatics and other Drexel students

Skills

Languages: Python, C#, C, C++, Bash/Shell Scripting, HTML/CSS, JavaScript

Databases: MySQL, PostgreSQL, OracleDB, MicrosoftSQL

Frameworks: FastAPI, Django, Flask, BeautifulSoup, .NET Core, .NET Framework, Entity Framework, jQuery

Tools: Docker, Git, Gitlab CI/CD, Jira, Confluence, CMake, YAML